

Part Number	Input Voltage	Max Current Rating	R _{DS_ON}
VL6102A	4.5V ~ 16V	15A	3mΩ

Description

The VL6102A is an active circuit protection device with an integrated FET. It is designed as a hot-swap circuit to limit currents, voltages to safe levels during fault conditions. The VL6102A also incorporates various functions to protect system against inrush current, over current, over voltage, reverse current, reverse polarity and short circuit failures.

The VL6102A incorporates a current monitoring function (IMON) of which voltage is produced using an external resistor connected from IMON to ground. Inrush current is limited by the slew rate at the output during start-up. The slew rate is controlled by connecting an external capacitor at the soft start (SS) pin.

The VL6102A is designed with programmable current limit. The current limit is programmed by an external resistor connecting from ISET pin to ground. An internal charge pump drives the gate of the internal MOSFET, allowing a FET with a very low on resistance of 3mΩ to turn on.

The VL6102A has fully integrated protection features including current limit, thermal shutdown, damaged MOSFET detection, over-voltage protection (OVP), and under-voltage protection (UVP).

The VL6102A is packaged in a 3x5mm QFN with 22 I/O pins.

Features

- Operating Input Range is from 4.5V to 16V
- Integrated 3mΩ Power FET capable of 15A maximum DC current
- Current monitoring function with ±3% Accuracy (6A < I_o < 15A)
- Programmable Current Limit
- PG Detector and FLTB Indication
- PG Assert Low at VIN = 0
- Fast Response (<200ns) for Short Circuit Protection
- Under-Voltage Lockout (UVLO)
- Over Temperature Protection
- Configurable Over-Voltage Lockouts with Hysteresis
- Input-to-Output Short-Circuit Detection
- External Soft Start
- Programmable LOADEN Blanking Time
- -40°C to 150°C Operation Junction Temperature
- 3 mm x 5 mm, 22 pin QFN package

Applications

- Hot Swap
- Networking
- Laptops
- PC Cards
- Disk Drives
- Servers

Typical Application Circuit

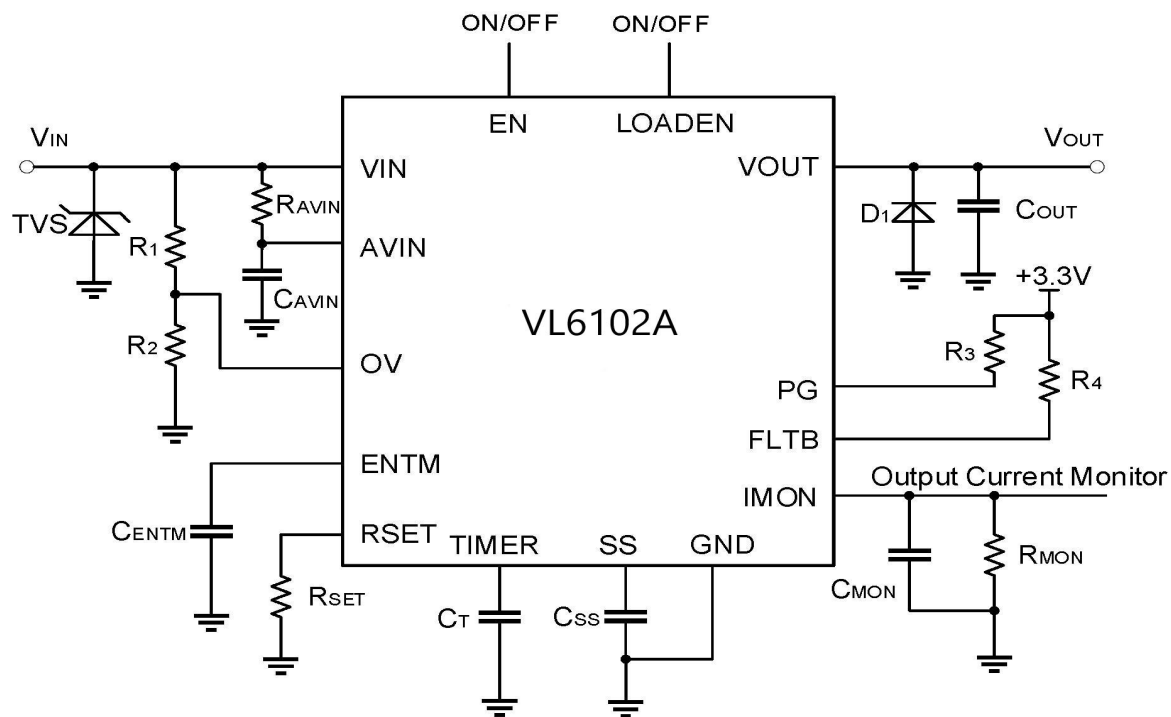


Figure 1. Application Circuit

Order Information

Part Number	Package	Shipping Method	Package Marking
VL6102A	QFN-22(3mm x 5mm)	5000u Tape & Reel	VL6102A

Top Marking (VL6102A)

VL6102A

XXXXXX

YYWW

VL: Vela Semiconductor prefix

6102A: First five digits of the part number

XXXXXX: Lot ID

Y: Year code

W: Week code

Package Reference

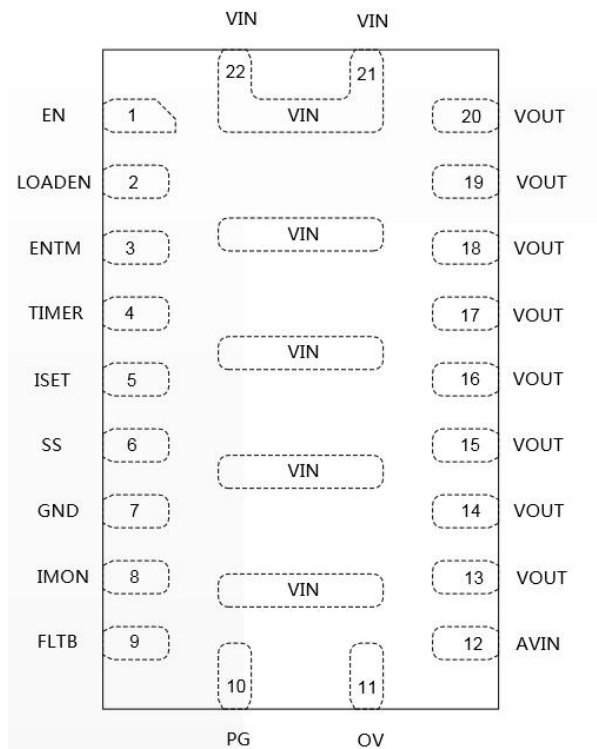


Figure 2. Top View

Absolute Maximum Ratings

Supply Voltage (V_{in}) -0.3V to 24V
 V_{out} Voltage (V_{out}).....-0.3V to 20V
All other pins.....-0.3V to 6.5V
Junction Temperature (T_j).....150°C
Storage Temperature.....-65°C to 150°C

Recommended Operating

Conditions

Supply Voltage (V_{in}).....4.5V to 16V
Max Output Current (I_{o_max})..... 15A
Junction Temperature (T_j)..... -40°C to 125°C

Thermal Ratings

Θ_{JA} Max.....46°C/W
 Θ_{JC} Max.....10°C/W

Note: (1) Stresses beyond those listed under “Absolute Maximum Ratings” may cause permanent damage to the device. These are stress ratings only; functional operation of the device at these or any other conditions beyond those indicated in the operational sections of the specifications is not implied. Exposure to absolute maximum rating conditions for extended periods may affect device reliability.

Pin Out

Pin Number	Name	Description
1	EN	Enable input. When used in conjunction with LOADEN it can turn on or off the main power device in the VL6102A. EN is pulled high internally.
2	LOADEN	Load enable input. Together with EN, it acts on the opening and closing of the main power transistor . After the fade time of LOADEN, it can also be used to turn off the power switch, but it cannot be turned on again.
3	ENTM	LOADEN blanking time set. Need to connect an external capacitor to set the LOADEN blanking time. Once EN is active, the timer starts and the LOADEN de-assertion is blanked. The switch is turned off in the presence of a fault or EN low condition, but LOADEN low during the blanking time does not turn off the power switch.
4	TIMER	Timer set. An external capacitor sets the hot-plug insertion time delay and fault time-out period.
5	ISET	Current limit set. Place a resistor from ISET to ground to set the value of the over current limit.
6	SS	Soft start. Place a capacitor from SS to GND to set output startup time. The internal circuit controls the slew rate of the output voltage during start-up. Floating SS pin sets the soft-start time to its minimum (1msec).
7	GND	Ground.
8	IMON	Output current monitoring. IMON supplies a voltage proportional to the current flowing through the power device. Placing a 10k Ω resistor (R_{MON}) to ground creates a 0V to 1.5V when the current is between 0A and 15A. Place a capacitor greater than 10nF in parallel with R_{MON} for the application.
9	FLT B	FLT B is an open-drain output that drives to ground when an over-current or a thermal shutdown happens. Pull FLT B up to an external power supply through a 10-100k Ω resistor.
10	PG	Open-drain power-good indicator for output. Pull PG up to an external power supply through a 10-100k Ω resistor. PG high means power is good.
11	OV	Pull OV pin high to turn off the internal power switch . Connect OV to an external resistive divider to set the over-voltage disable threshold.
12	AVIN	Internal power supply for VCC regulator. Connect a 10 Ω , 0603 resistor from VIN to AVIN and a 10 μ F bypass capacitor to GND to ensure full operation in the event that VIN collapses upon a strong short from VOUT to GND.
13-20	VOUT	Output. VOUT is the output voltage programmed by the IC. Need to place a Schottky diode from VOUT and GND to minimize negative voltage spike.
21,22	VIN	Main power input for the IC.

Functional Block Diagram

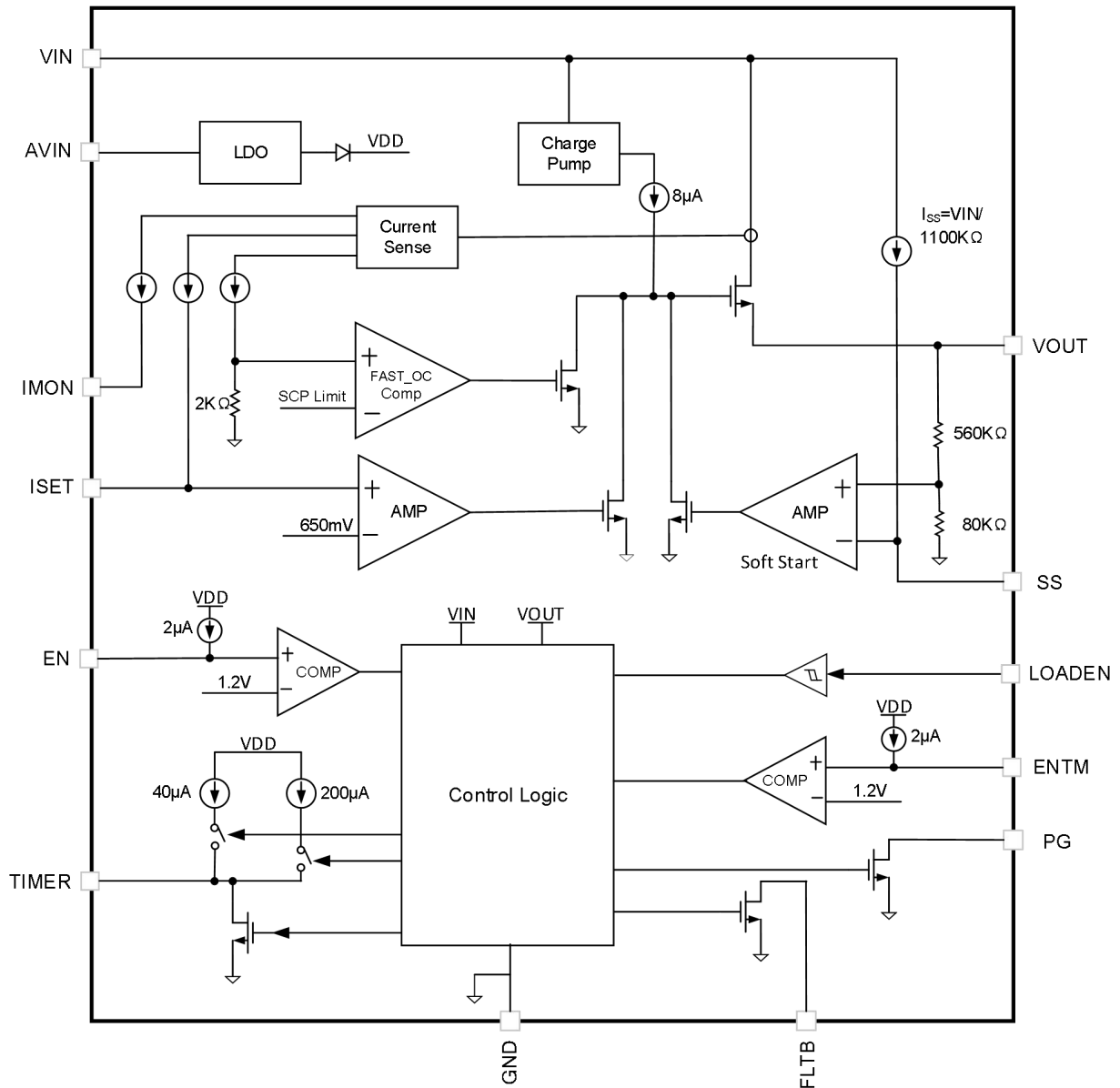


Figure 3. Functional Block Diagram

Electrical Characteristics

$V_{IN} = 12V$, $R_{SET} = 10.2k$, $C_{OUT} = 246\mu F$, $T_J = 25^\circ C$, unless otherwise noted.

Parameters	Symbol	Condition	Min	Typ	Max	Units
Supply Current						
Quiescent current	I _Q	EN = high, No Load		1.9		mA
		Fault latch-off		1.9		mA
		EN = 0, VIN = 16V			700	μA
Power FET						
On resistance	R _{DS(ON)}	T _J = 25°C		3	4	mΩ
		T _J = 85°C		3.8		mΩ
Off-state leakage current	I _{OFF}	VIN = 24V, EN = 0V			1	μA
Maximum continuous current ⁽²⁾	I _{OUT_MAX}		15			A
Thermal Shutdown						
Shutdown temperature ⁽²⁾	t _{STD}			145		°C
VIN Under-Voltage Protection(UVLO)						
VINUVLO threshold	VIN _{UVLO}	VIN _{UVLO} rising threshold		4.05	4.4	V
VINUVLO hysteresis	VIN _{UVLOHYS}			0.32		V
LOADEN						
Low-level input voltage	V _L				0.9	V
High-level input voltage	V _H		2.4			V
Soft Start(SS)						
SS pull-up current	I _{SS}	V _{SS} = 0V		12		μA
Current Limit						
Current limit at normal operation	I _{Limit_NO}	R _{SET} = 10.2k		12.6		A
Current limit response time ⁽²⁾	t _{CL}			40		μs
Secondary current limit ⁽²⁾	I _{LimitH}	Regardless of R _{SET}		37		A
Short-circuit protection response time ⁽²⁾	t _{SC}			200		ns
Output Current Monitor						
Gain of current sense amplifier	A _{IMON}	3A < I _{OUT} < 6A		10		μA/A
		6A < I _{OUT} < 15A		10		μA/A
Max IMON voltage	V _{IMON}			4.47		V
Timer						
Upper threshold voltage	V _{TMRH}			1.21		V
Insertion delay charge current	I _{INSERT}			40		μA
Fault detection charge current	I _{FLTD}			200		μA
Discharge R _{DS(ON)}	R _{FLTE}	I _{OUT} < I _{Limit}		48		Ω
LOADEN Blanking Time(ENTM)						
Upper threshold voltage	V _{ENTMRH}			1.2		V

Charge current	I_{ENTMCC}			2		μA
Enable						
Rising threshold	V_{ENRS}			1.18		V
Hysteresis	V_{ENHYS}			210		mV
Over-Voltage(OV)						
OV threshold	V_{OV_TH}	V_{OV} rising		1.2		V
OV threshold hysteresis	V_{OV_HYS}	V_{OV} falling		110		mV
Fault Bar(FLT B) / Power Good(PG)						
Low-level output voltage	V_{OL}	Sink current 1mA			0.2	V
Fault bar off-state leakage current	I_{FLT_LKG}	$V_{FLT B} = 3.3V$			1	μA
Fault bar propagation delay	t_{PDE}	ISET stepped to 1V to FLT B pull-down		4		μs
Power good rising threshold ⁽²⁾	PG_{Vth_Hi}			90%		VIN
Power good falling threshold	PG_{Vth_Lo}			75%	80 %	VIN
Power good off-state leakage current	I_{PG_LKG}	$V_{PG} = 3.3V$			2.5	μA
PG low-level output voltage	V_{OL_100}	VIN = 0V, pull up to 3.3V through 100k Ω resistor		710		mV
	V_{OL_10}	VIN = 0V, pull up to 3.3V through 10k Ω resistor		820		mV

Note:

2) Guaranteed by design.

Typical Performance

$V_{IN} = 12V$, $C_{OUT} = 246\mu F$, $C_{ENTM} = 1\mu F$, $C_T = 220nF$, $C_{SS} = 47nF$, $R_{SET} = 6.8 k\Omega$, $T_A = +25^\circ C$, unless otherwise noted.

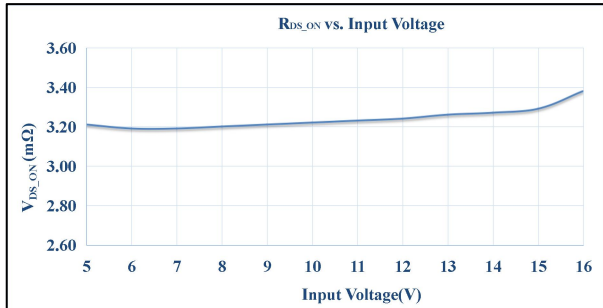


Figure 4. R_{DS_ON} vs Input Voltage
 $I_{OUT} = 1A$

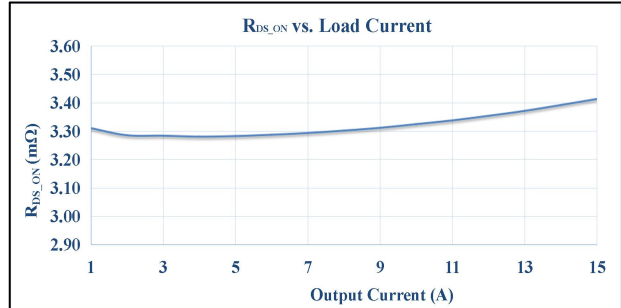


Figure 5. R_{DS_ON} vs Load Current
 $V_{IN} = 12V$

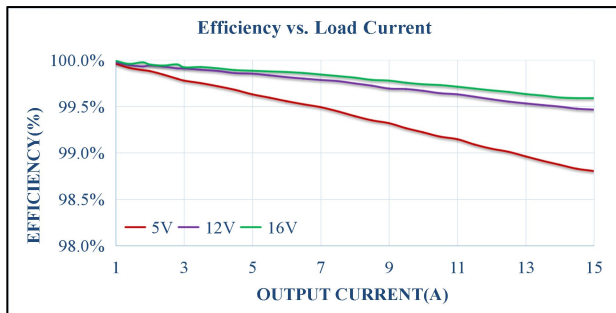


Figure 6. Efficiency vs Load Current
 $V_{IN}=5V/12V/16V$, $I_{OUT} = 1\sim 15A$

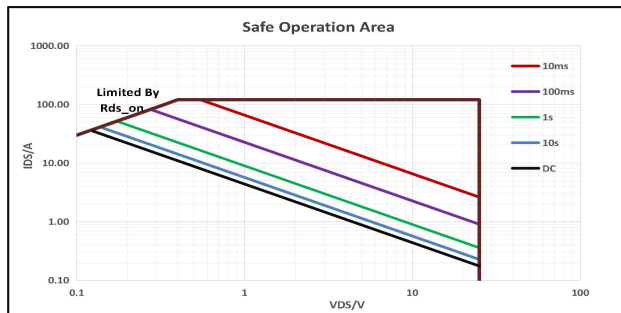


Figure 7. Safe Operation Area

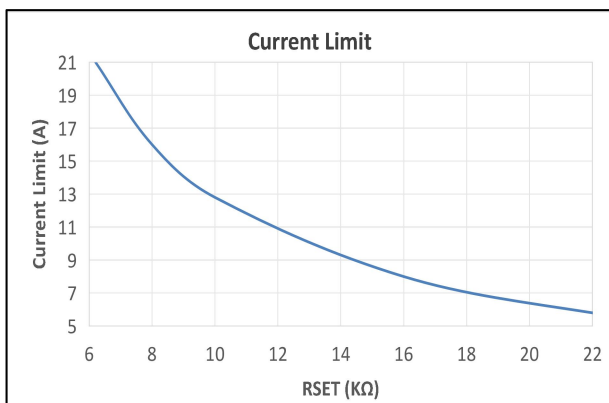


Figure 8. Current Limit vs R_{SET}
Current Limit > 7A

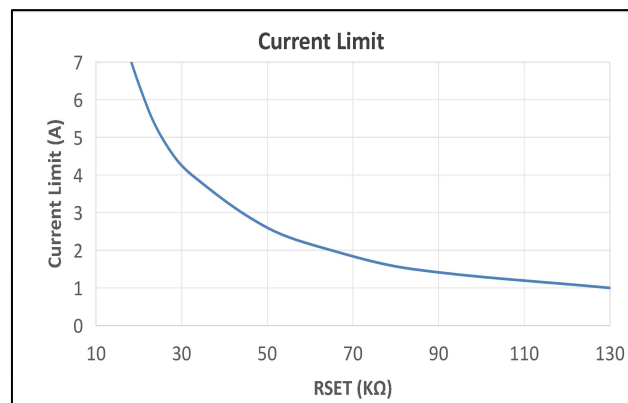


Figure 9. Current Limit vs R_{SET}
Current Limit < 7A

$V_{IN} = 12V$, $C_{OUT} = 246\mu F$, $C_{ENTM} = 1\mu F$, $C_T = 220nF$, $C_{SS} = 47nF$, $R_{SET} = 6.8 k\Omega$, $T_A = +25^\circ C$, unless otherwise noted.

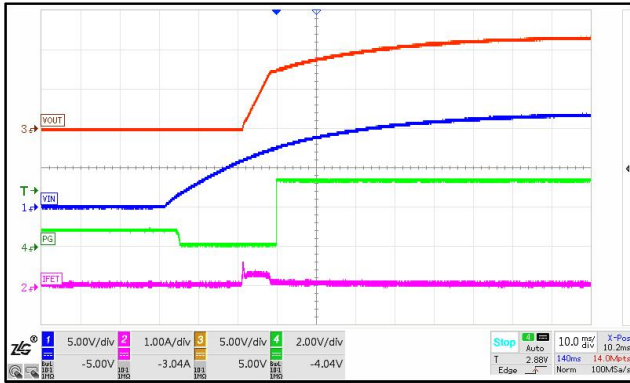


Figure 10. Start Up through VIN
 $I_{OUT}=0A$

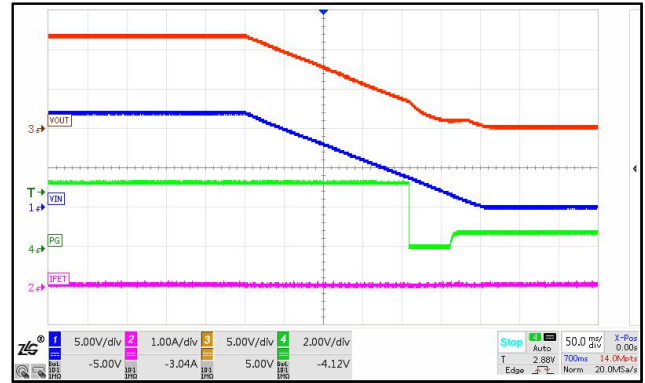


Figure 11. Shut Down through VIN
 $I_{OUT}=0A$

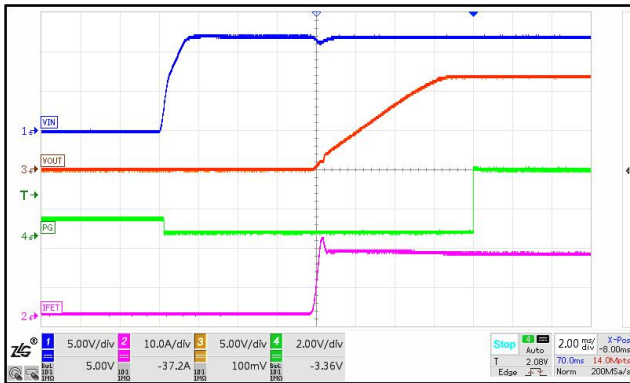


Figure 12. Start Up through VIN
 $I_{OUT}=15A$

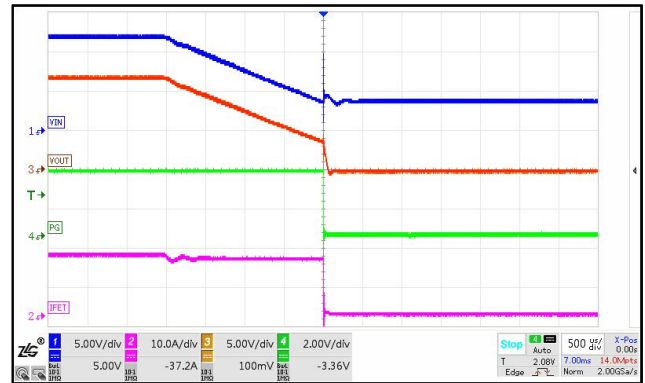


Figure 13. Shut Down through VIN
 $I_{OUT}=15A$

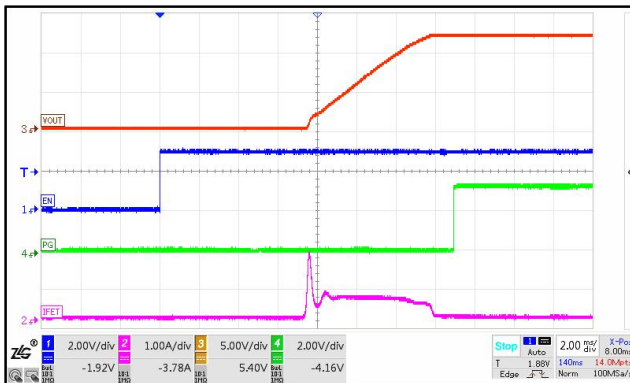


Figure 14. Start Up through EN
 $I_{OUT}=0A$

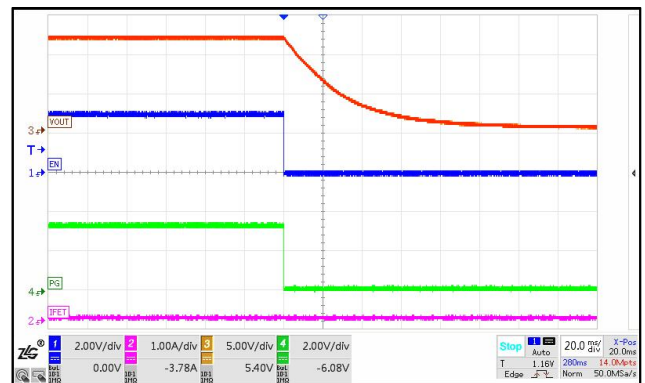


Figure 15. Shut Down through EN
 $I_{OUT}=0A$

$V_{IN} = 12V$, $C_{OUT} = 246\mu F$, $C_{ENTM} = 1\mu F$, $C_T = 220nF$, $C_{SS} = 47nF$, $R_{SET} = 6.8 k\Omega$, $T_A = +25^\circ C$, unless otherwise noted.

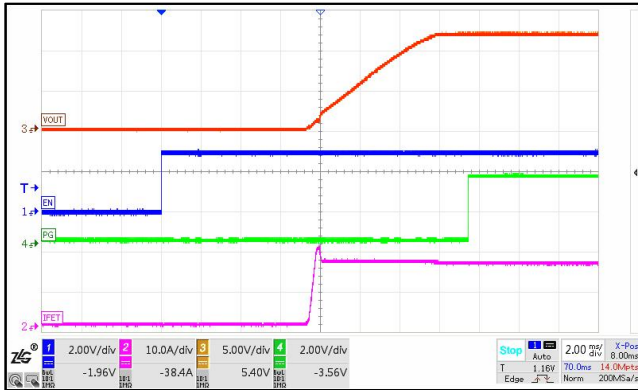


Figure 16. Start Up through EN
 $I_{OUT}=15A$

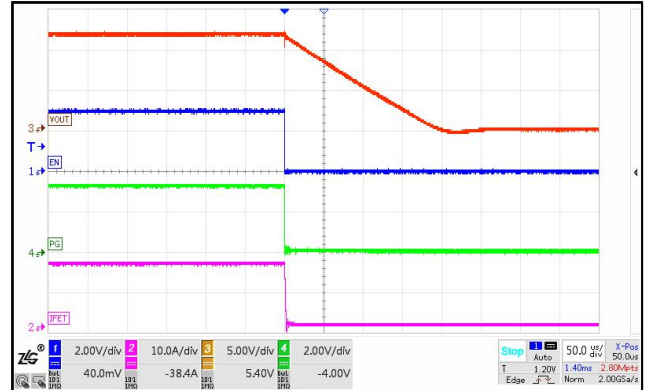


Figure 17. Shut Down through VIN
 $I_{OUT}=15A$

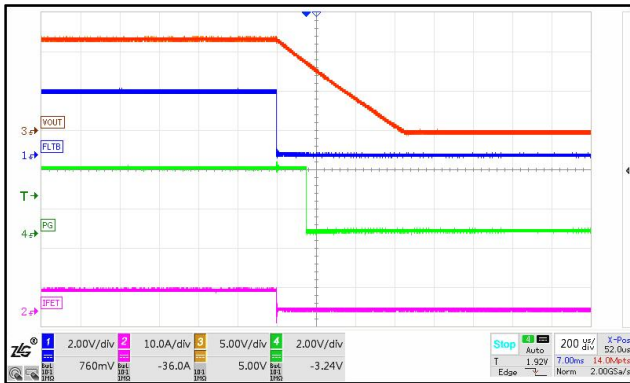


Figure 18. Thermal Shut Down
 $I_{OUT}=15A$

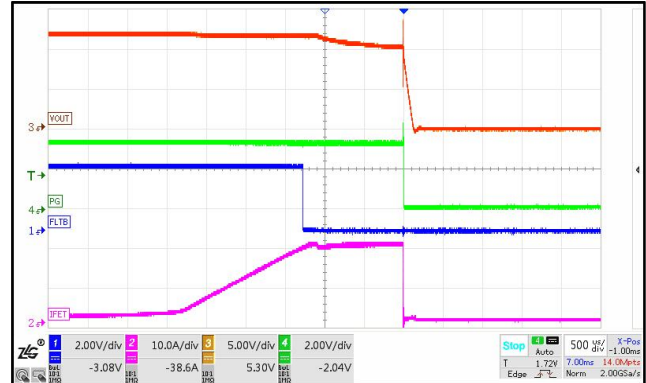


Figure 19. OCP
 $ILIMIT=19A$

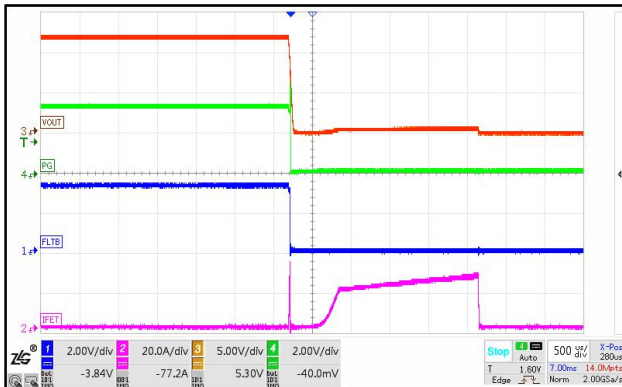


Figure 20. S_{CP} Entry
 $ILIMIT=19A$

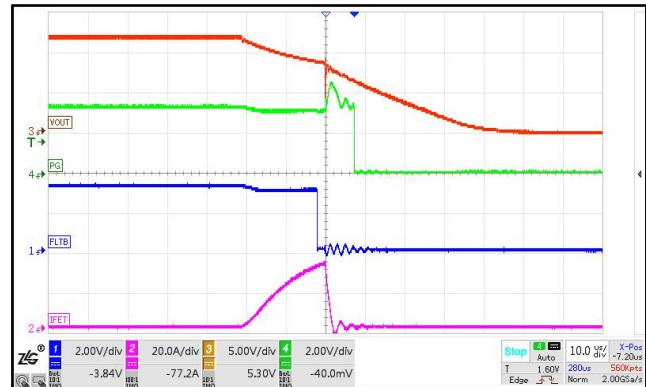


Figure 21. S_{CP} Entry Detail
 $ILIMIT=19A$

$V_{IN} = 12V$, $C_{OUT} = 246\mu F$, $C_{ENTM} = 1\mu F$, $C_T = 220nF$, $C_{SS} = 47nF$, $R_{SET} = 6.8 k\Omega$, $T_A = +25^\circ C$, unless otherwise noted.

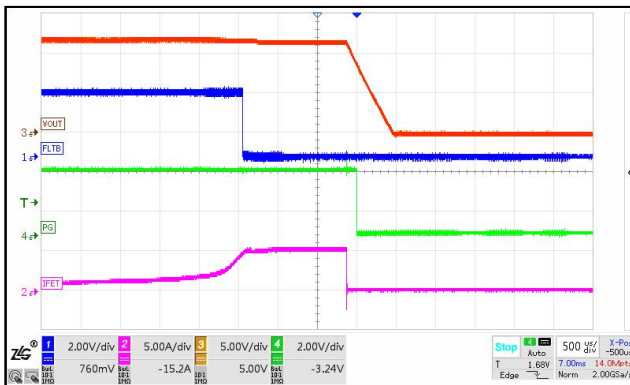


Figure 22. OCP
ILIMIT=5A

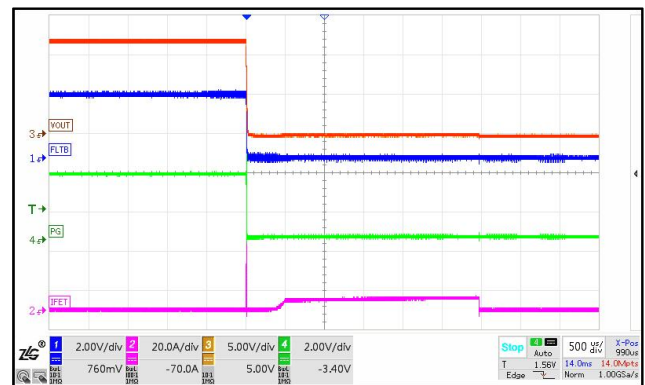


Figure 23. SCP Entry
ILIMIT=5A

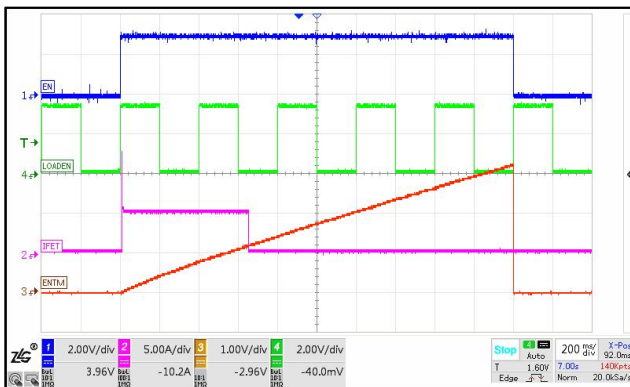


Figure 24. EN and LOADEN Logic
 $I_{OUT}=5A$

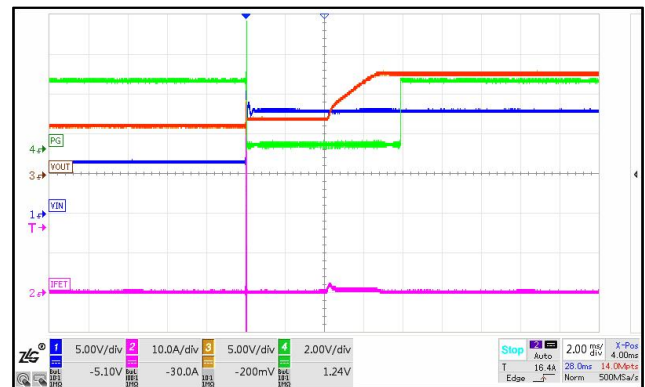


Figure 25. LINE SPIKE
ILIMIT=19A

Functional Block Diagram

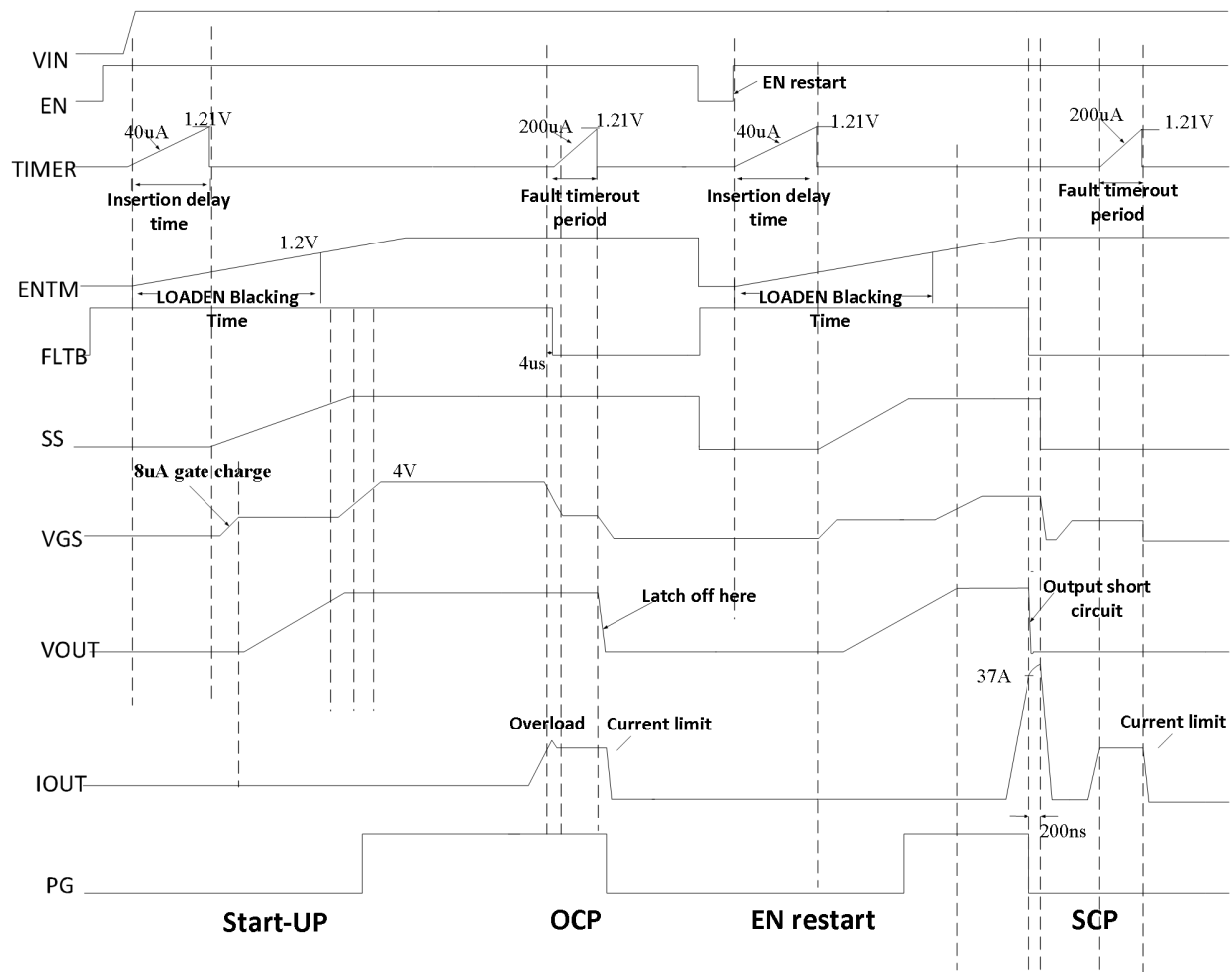


Figure 26. Over-Current Protection

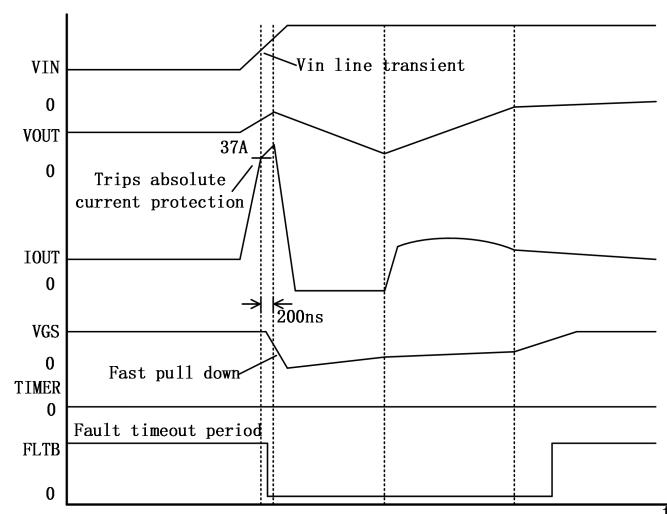


Figure 27. VIN Line Transient Protection

DETAIL APPLICATION DESCRIPTION

The VL6102A is designed to limit the inrush current of the load generated when inserting a circuit board into a live backplane power supply, thereby limiting the voltage drop of the backplane and the dV/dt of the load voltage. The VL6102A integrates monitoring functions for input voltage, output voltage, output current, and IC temperature, without the need for external sampling power resistor, power MOSFET, and thermal sensing device.

Current Limit Design

The VL6102A provides stable current limiting values, which can be programmed through an external resistor. Once the device current reaches the current limiting threshold, the fault timer starts counting. If the output current drops below the current limiting threshold before the fault timer expires, the VL6102A resumes normal operation. Otherwise, if the current limiting time exceeds the fault timeout, the power FET is latched off. When the IC reaches the over current or over temperature threshold, the FLTB signal is pulled down, and there is a delay of approximately 4μs to indicate a fault. The current limiting value during normal operation is a function of the external current limiting resistance. The capacitance value of C_T can be determined by formula (1):

$$I_{LIMIT}(A) = \frac{1.3(V)}{R_{SET}(k\Omega)} \times 10^5 \quad (1)$$

Where R_{SET} is the value of the external current limiting resistor.

Short-Circuit Protection (SCP)

In the situation where there is a hard short on the load, the current limit design is not fast enough to react due to the control loop response. If the current reaches the secondary current limiting value of 37A, the internal fast shutdown circuit acts to turn off the power MOSFET through a 100mA pull-down gate discharge current (see Figure 26). Limiting input voltage drop is accomplished by limiting the peak current flowing through the device. The total response time for short circuit protection is less than 200ns.

When short circuit protection (SCP) is triggered, the IC restarts to determine if an overload condition exists. If the short circuit is caused by a transient in the input line, the IC can function normally. If a real short circuit occurs, the IC is completely latched off (see Figure 26 and Figure 27). Once the current limit of 37A is reached, the FLTB is set low and remains deasserted until the short circuit condition is removed.

Fault Timer and Restart

When the current reaches the current limit threshold, the internal 200μA fault timer current source charges the external capacitor (C_T) of TIMER. If the over current protection state

disappears before TIMER reaches 1.21V, VL6102A will return to normal operating mode and release TIMER immediately after the over current protection is eliminated. If it still shows over current protection status after TIMER reaches 1.21V, then turn off the power FET. In addition, the fault timing capacitor also determines the insertion delay timing during startup. The capacitance value of C_T can be determined by formula (2):

$$C_T(\text{nF}) = \frac{200(\mu\text{A}) \times t_{\text{fault}}(\text{mS})}{1.21(\text{V})} \quad (2)$$

Power Good (PG)

Power Good (PG) is used to indicate whether the output voltage is within the normal range corresponding to the input voltage. PG is also an open drain FET structure. Pull PG up to an external power source through a 10-100 k Ω resistor. When starting up, the PG is forced low, indicating that the system is in an OFF state. When the IC meets the following conditions ($V_{\text{OUT}} > 90\% * V_{\text{IN}}$, $V_{\text{GS}} > 3\text{V}$, $V_{\text{OUT}} > V_{\text{IN}} - 0.8\text{V}$), the PG signal is pulled up to a high level, and the system can operate at full power.

When $V_{\text{OUT}} < 75\% * V_{\text{IN}}$, PG deasserts. When EN falls below its threshold, PG output drops to a low level. Without input, PG remains deasserted in the presence of pull-up power.

Fault Bar (FLT B)

Fault Bar (FLT B) is an open drain output used to indicate whether a fault has occurred. Pull the FLT B up to an external power source through a 10-100 k Ω resistor. When the IC is in over current, over temperature, and/or MOSFET is short circuited condition before power on, the fault output is forced low after a delay of about 8 μs . If the IC experiences a short circuit and reaches the secondary current limit of 37A, the FLT B is immediately deasserted. After the VL6102A resumes normal operation, FLT B returns to high level. This means that the output voltage is higher than the voltage set by the PG rising threshold, and the power FET is fully turned on ($V_{\text{GS}} > 3\text{V}$).

External Supply for PG and FLT B

PG and FLT B require an external power supply. The open-drain output of PG can work from the external pull-up voltage, even when $V_{\text{IN}} = 0$ and EN is disabled. Use a 10-100 k Ω pull-up resistor for PG and FLT B.

Startup Sequence

For hot swap applications, the input of VL6102A withstands voltage spikes/transients during hot swapping. These voltage transients are caused by parasitic inductance and input capacitance in the input circuit. In order to stabilize the input voltage, it is necessary to use an insertion delay before the main power FET is turned on. When the input voltage reaches

the UVLO threshold, TIMER will charge the external capacitor (C_T) through a $40\mu\text{A}$ constant current source (see Figure 28). When the TIMER voltage reaches 1.21V , the insertion delay ends. The capacitance value of C_T can be determined by formula (3):

$$C_T(\text{nF}) = \frac{40(\mu\text{A}) \times t_{\text{delay}}(\text{mS})}{1.21(\text{V})} \quad (3)$$

In addition, the external capacitor of TIMER will also determine the timing of the fault delay. After TIMER reaches 1.21V , the $8\mu\text{A}$ current source begins to charge the power FET gate, and the gate voltage rises. Meanwhile, the TIMER voltage discharges until it reaches zero. Once the gate voltage reaches its threshold (V_{GSTH}), the output voltage begins to rise. The rise time is determined by the soft start capacitor.

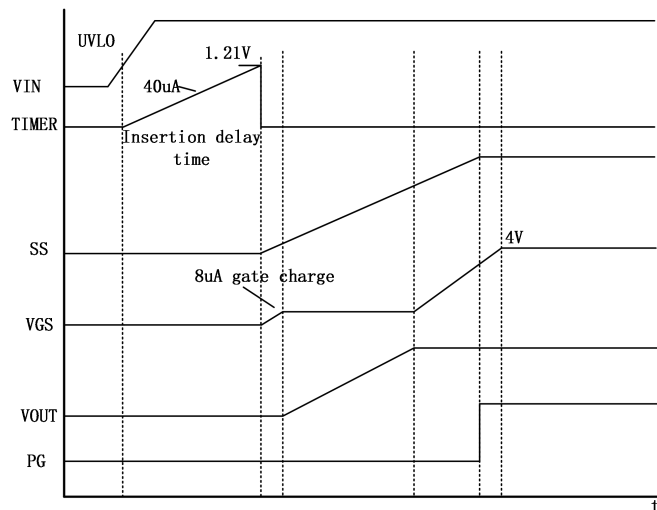


Figure 28. Startup Sequence

Soft Start (SS)

The soft start time is determined by the capacitance connected to the SS pin. After the insertion delay time ends, a constant current source directly proportional to the input voltage inside the IC charges the external capacitor, causing its voltage to rise at a corresponding slope, and the output voltage will rise at the same slope. To generate a fast ramp-up voltage, leave SS pin float. An $8\mu\text{A}$ current source is used to pull up the gate of the power FET. The gate charging current controls the rise time of the output voltage, and the minimum soft start time is approximately $750\mu\text{s}$. The capacitance value of SS can be calculated according to the following formula (4):

$$C_{\text{SS}}(\text{nF}) = \frac{8 \times t_{\text{ss}}(\text{mS})}{1.1(\text{M}\Omega)} \quad (4)$$

If the load capacitance is very large, the current used to maintain the preset soft start time

exceeds the current limit value. In this case, the rise time is controlled by the load capacitance and current limit setpoint.

Enable (EN) and Load Enable (LOADEN)

EN and LOADEN are used to control the on/off of VL6102A .

During the LOADEN blanking period, the IC can be activated when EN=1. After the LOADEN blanking time ends, the device must be turned on with EN=1 and LOADEN=1. At any time, EN=0 can be used to turn off the chip. Once the IC is latched off, a restart of EN or VIN is required to restart the IC.

Note: After the blanking time, LOADEN can also be used to turn off the power switch, but it cannot be used to turn on again.

EN is pulled up to a high level by a 2 μ A internal pull-up power supply. Once the chip starts, the insertion delay timer will start. When the insertion delay time ends, internal 8 μ A's current source charges the power FET gate. After charging for about 1ms, V_{GS} reaches the threshold, and then the output voltage increases with SS slope.

LOADEN Blanking Time

If EN is asserted, the LOADEN can not be de-asserted during the blanking time window. This blanking time can be programmed. During the startup process, all fault functions are active (see Figure 29). If a fault is detected, the power switch is turned off. However, if LOADEN is deasserted during the blanking period, the switch cannot be turned off. After the blanking time, LOADEN returns to normal operation.

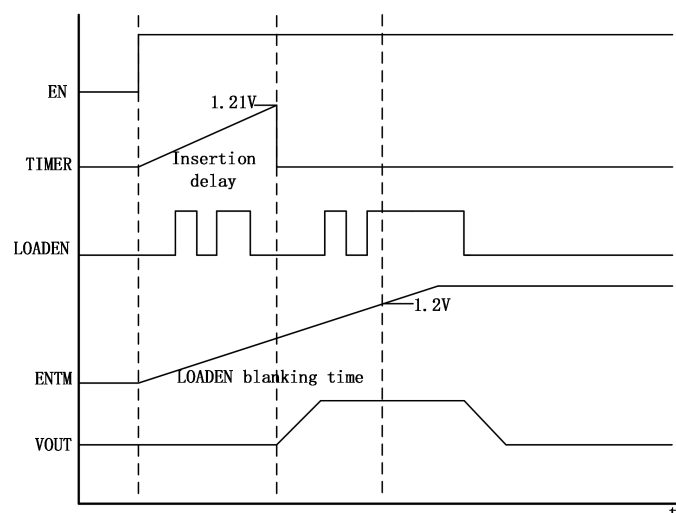


Figure 29. LOADEN Blanking Time

The blanking time can be set by a capacitor connected to the ENTM pin. This capacitance can be calculated according to the following formula (5):

$$C_{ENTM}(\text{nF}) = \frac{t_{LDNB}(\text{mS}) \times 10^{-6}}{1.2(V)} \quad (5)$$

Where t_{LDNB} is the LOADEN blanking time, C_{ENTM} is the blanking time capacitor on ENTM pin.

Pass FET Short-Circuit Detection

The VL6102A can detect a main pass FET during power up by treating an output voltage exceeding $V_{IN} - 0.8V$ during start-up as a short on the MOSFET.

FLTB goes low to indicate a fault condition and the power switch is latched off.

Once V_{OUT} is less than or equal to $V_{IN} - 0.8$, the part starts up normally.

Internal VCC LDO

The VL6102A has an internal 4V linear sub-regulator that steps down the input voltage to generate a 4V bias supply that powers low-voltage circuitries. The LDO regulator is enabled when V_{IN} exceeds its UVLO threshold and EN is high.

Input Supply for Internal LDO (AVIN)

AVIN is the power supply for the internal VCC sub-regulator. Connect a 10Ω resistor from V_{IN} to AVIN and a $10\mu F$ bypass capacitor to GND to guarantee full operation in the event V_{IN} collapses during a strong short from V_{OUT} to GND. V_{IN} has UVLO protection, but AVIN does not have UVLO protection. Do not use AVIN alone to power off the VL6102A. The AVIN supply current is $500\mu A$, typically.

Over-Voltage Lockout (OVLO)

The VL6102A monitors the supply voltage through OV pin for input over-voltage conditions. An external resistor divider from V_{IN} to OV provides flexibility for setting the over-voltage lockout threshold. When the voltage on OV pin exceeds 1.2V, the internal MOSFET is shutdown, and the output is disabled. When the voltage on OV pin falls below $1.2V - V_{OV_HYS}$, internal MOSFET is turned on again and the output ramps up with a pre-determined soft-start time.

Under-Voltage Lockout (UVLO)

If the supply (input) falls below the under-voltage lockout (UVLO) threshold, the output is disabled and PG goes low. When the supply exceeds the UVLO threshold without exceeding the OV threshold, the output is enabled.

Output Current Monitoring (IMON)

IMON provides a current value that is proportional to the output current (the current flowing through the power device). The gain of the current amplifier is $10\mu A/A$, which means that

IMON provides 10 $\mu\text{A/A}$ current for each main FET amplifier. When the current range of MOSFET is 0A to 20A, connecting a 10k Ω resistor to GND generates a voltage of 0V to 2V. In the application circuit, it is necessary to parallel a capacitor of 10nF or more with the R_{MON} .

TYPICAL APPLICATION CIRCUITS

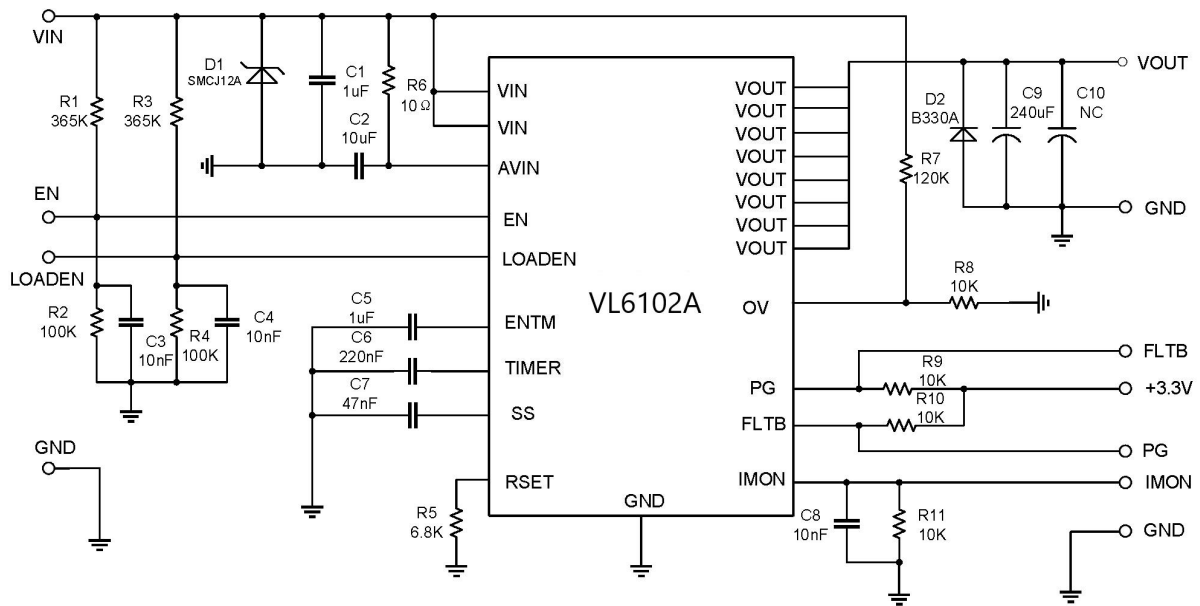


Figure 30. Over-Current Limit > 7A Application

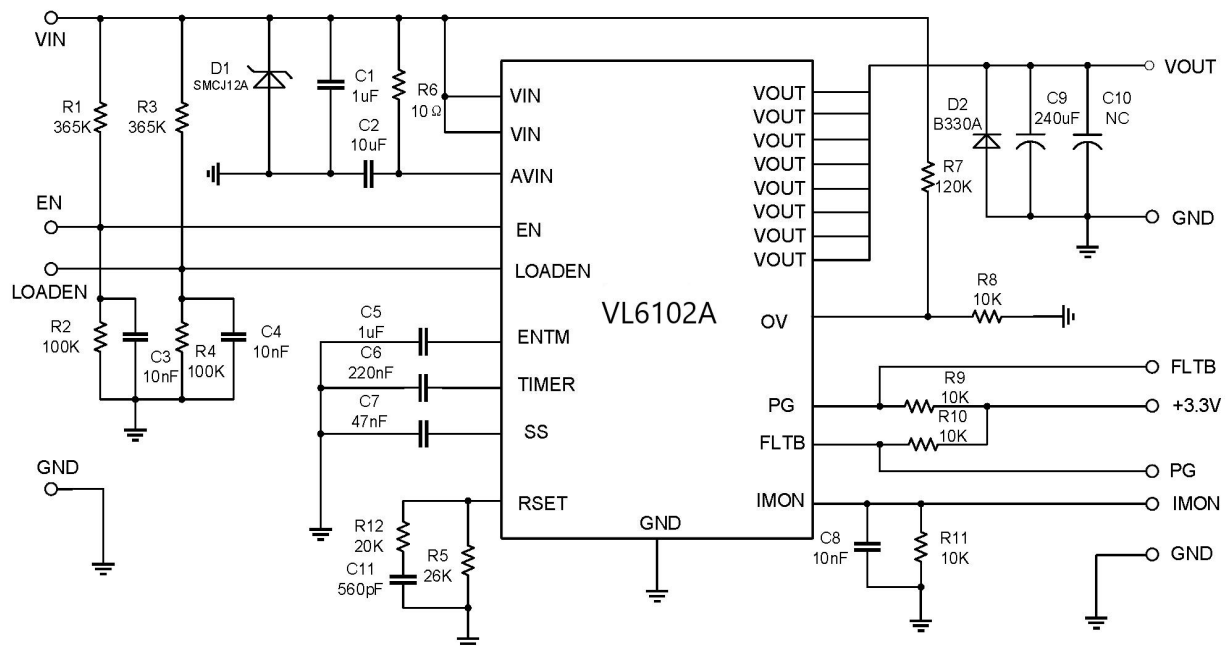


Figure 31. Over-Current Limit < 7A Application

PCB Layout Guidelines

PCB layout is critical to the stable operation of the circuit. For optimized performance, please follow the PCB layout recommendations below.

1. A high current path should be placed between the input and output terminals of the circuit board, with the current return path parallel to minimize the current loop therefore reduce loop inductance.
2. Place a transient voltage suppressor diode (TVS) at VIN. And place TVS as close as possible to VIN.
3. Connect the GND of VL6102A to a small ground island as a reference for all signal grounding of the IC. The signal GND can be connected to the main power GND of the system through a single point of contact.
4. Place the Schottky diode close to the VOUT and GND to absorb negative voltage transients when the power FET is turned off.
5. Make sure that the input decoupling capacitor on VIN has the shortest trace to VIN and GND.
6. Place the output capacitors as close to VL6102A as possible to minimize the impact of PCB parasitic inductance.
7. Add as many through holes on the thermal pad to achieve better heat dissipation performance.

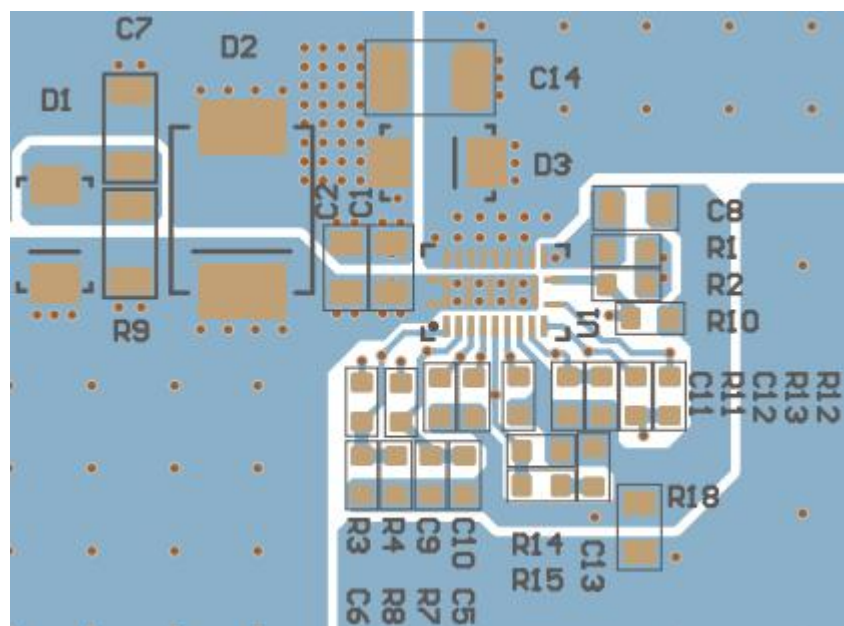
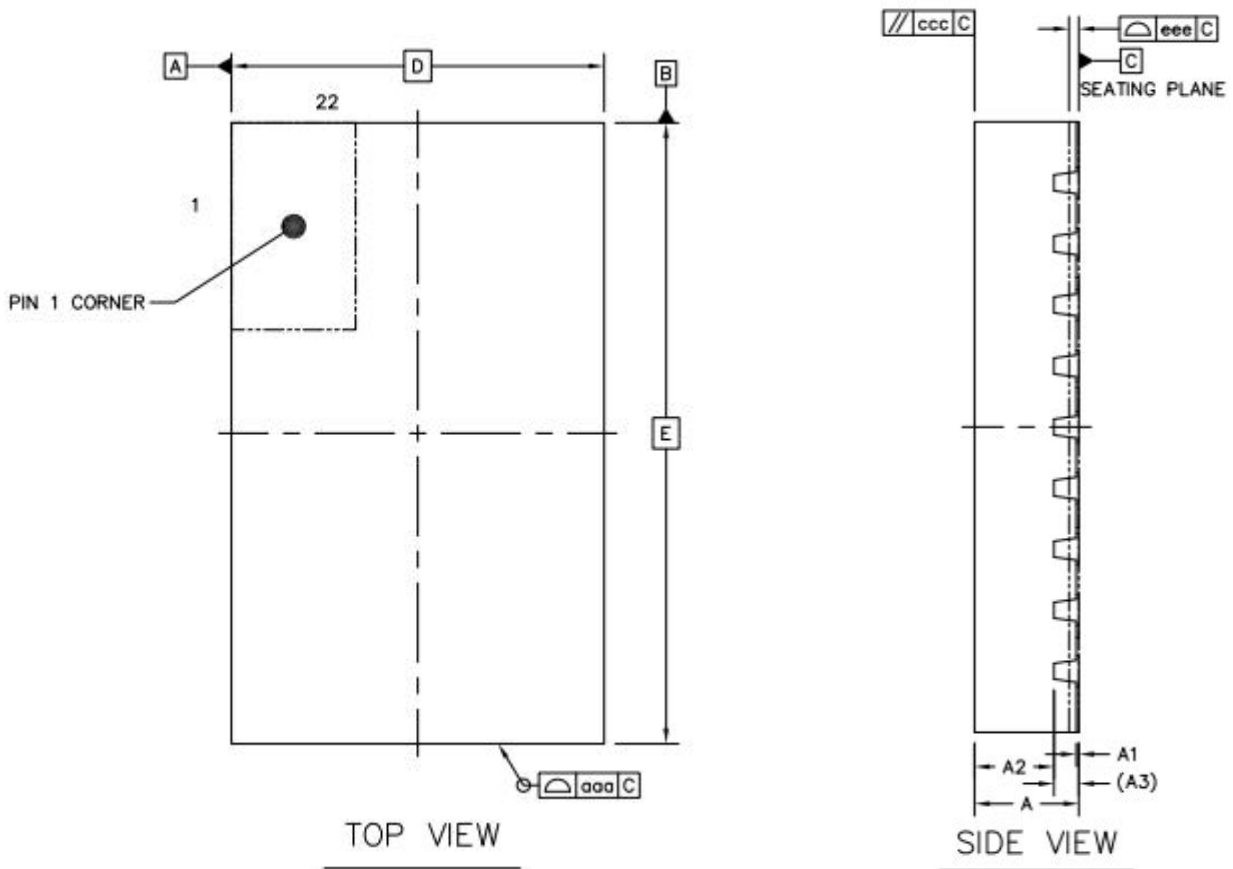
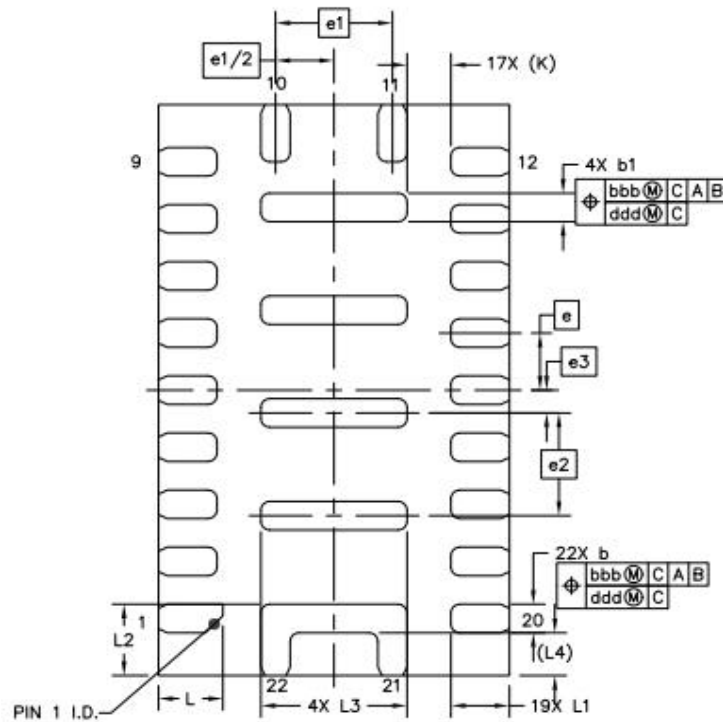


Figure 32. Recommended PCB layout

Unit(mm)

Package Dimension



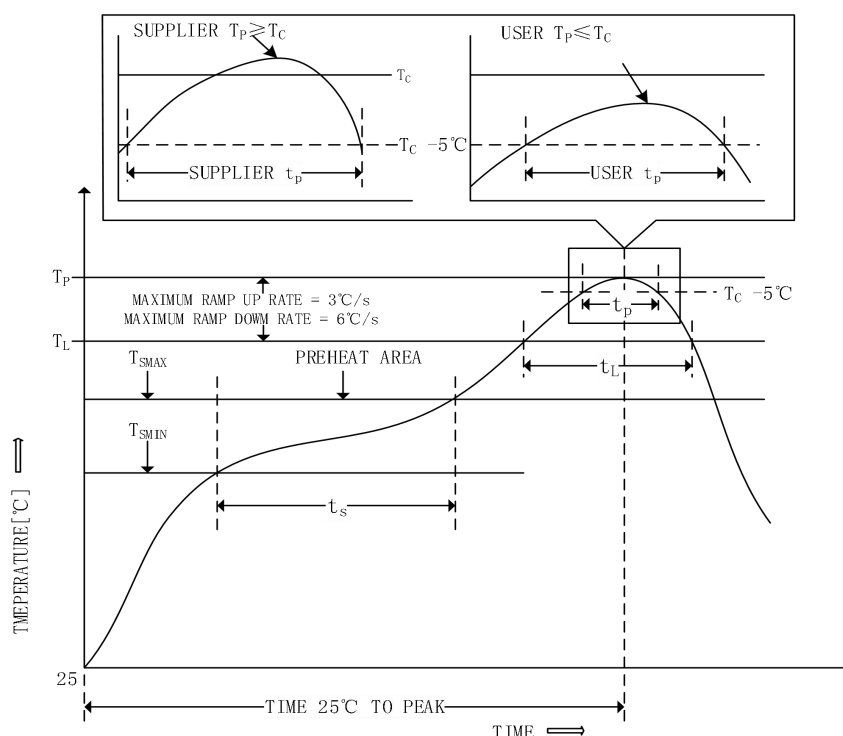

BOTTOM VIEW

		SYMBOL	MIN	NOM	MAX
TOTAL THICKNESS		A	0.8	0.85	0.9
STAND OFF		A1	0	0.02	0.05
MOLD THICKNESS		A2	---	0.65	---
L/F THICKNESS		A3	0.203 REF		
LEAD WIDTH		b	0.2	0.25	0.3
		b1	0.15	0.25	0.35
BODY SIZE	X	D	3 BSC		
	Y	E	5 BSC		
LEAD PITCH		e	0.5 BSC		
		e1	1 BSC		
		e2	0.9 BSC		
LEAD CENTER TO PKG CENTER		e3	0.2 BSC		
LEAD LENGTH		L	0.45	0.55	0.65
		L1	0.4	0.5	0.6
		L2	0.525	0.625	0.725
		L3	1.15	1.25	1.35
LEAD EDGE TO PKG EDGE		L4	0.375 REF		
LEAD EDGE TO LEAD EDGE		K	0.375 REF		
PACKAGE EDGE TOLERANCE		aaa	0.1		
MOLD FLATNESS		ccc	0.1		
COPLANARITY		eee	0.08		
LEAD OFFSET		bbb	0.1		
		ddd	0.05		

Reflow Specification

Qualification Reflow: The VL6102A was qualified in accordance with IPC/JEDEC J-STD-020D.01. This standard classifies proper packaging, storage and handling in order to avoid subsequent thermal and mechanical damage during the solder reflow attachment phase of PCB assembly.

The qualification preconditioning process specifies a sequence consisting of a bake cycle, moisture soak cycle (in a temperature humidity oven), and three consecutive solder reflow cycles, followed by functional device testing.



Production Reflow:

PROFILE FEATURE	SN - PB EUTECTIC ASSEMBLY	PB-FREE ASSEMBLY
Peak package body temperature (TP)	For users, TP must not exceed Tc(235°C).For suppliers, TP must equal or exceed Tc(235°C).	For users, TP must not exceed Tc(260°C).For suppliers, TP must equal or exceed Tc(260°C).

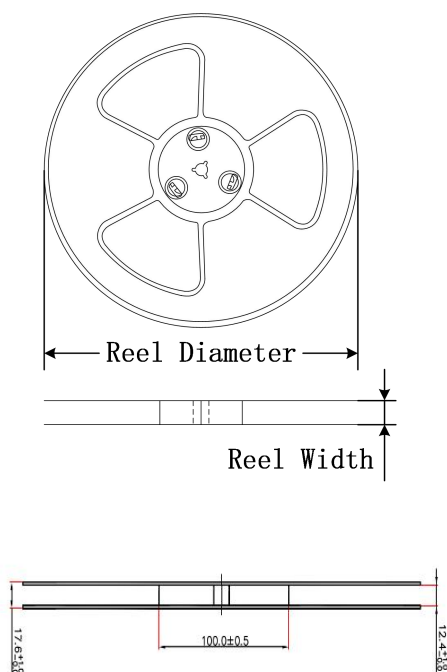
Storage Specifications

The storage specification of the VL6102A conforms to IPC/JEDEC J-STD-020D.01 Moisture Sensitivity Level (MSL) 3.

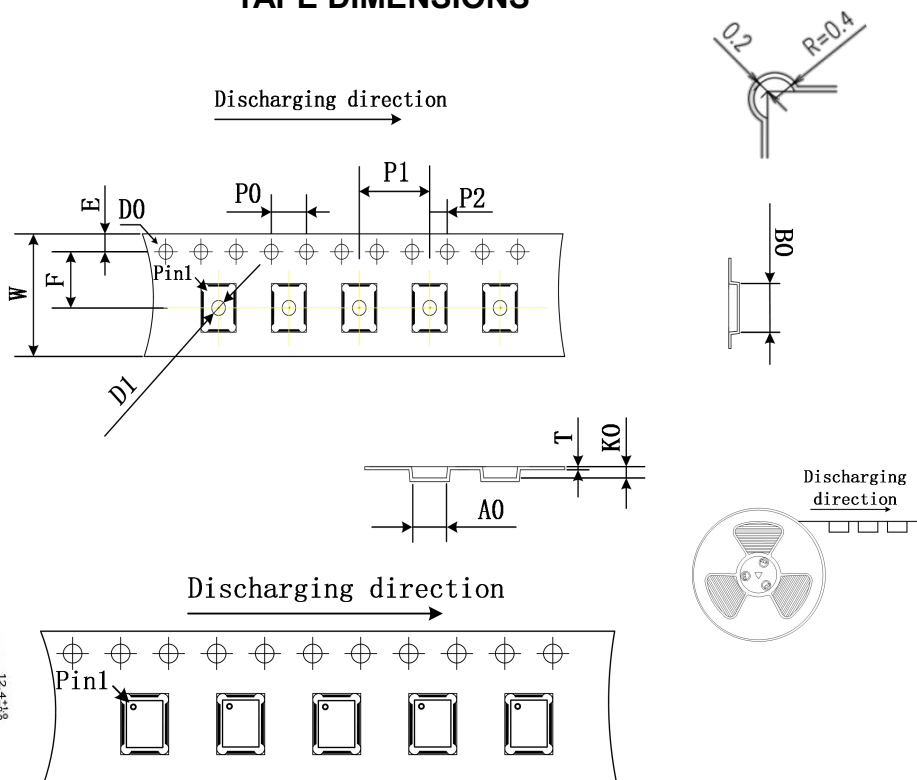
After opening moisture-sealed bag	168 hours -- Storage conditions: ambient ≤30°C at 60%RH
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TAPE AND REEL INFORMATION

REEL DIMENSIONS



TAPE DIMENSIONS



KEY PARAMETER LIST OF TAPE AND REEL

Package Type	Reel Diameter (inch)	Reel Width (mm)
QFN-22(3 x 5)	13"	17.6 ±1.0

NOTE:

- 1) .The accumulative error of any 10 chain holes shall not exceed $\pm 0.2\text{mm}$
- 2) . The non-parallelism of 100mm long carrier band in the carrier band length direction shall not exceed 1mm.
- 3) . Material: black, PS material.
- 4) .All dimensions meet the requirements of EIA-481-D.
- 5) . This picture is for reference only. Refer to the actual product.

Unit(mm)

EIA	W	E	F	A0	B0	K0	P0	P1	P2	D0	D1	T
DIM	12 ±0.3	1.75 ±0.1	5.5 ±0.05	3.25 ±0.1	5.25 ±0.1	1.13 ±0.1	4.00 ±0.05	8.00 ±0.1	2.00 ±0.05	1.5 ±0.1	1.5 ±0.25	0.25 ±0.02